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ESTIMATING THE DETERMINANTS OF TAXPAYER COMPLIANCE WITH EXPERIMENTAL DATA***

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ABSTRACT

The fundamental difficulty in empirical work on taxpayer compliance is the absence of detailed and reliable information on individual compliance choices. This paper uses data from laboratory experiments to estimate individual responses to tax, penalty, and audit rate changes, as well as to changes in government expenditures. The empirical results confirm some (although not all) theoretical predictions, and compare qualititatively with other empirical work. Taxpayer reporting increases with greater audit and penalty rates; however, these responses are not large. Compliance is also greater when individuals face a lower tax rate and when they receive something for their taxes.

SINCE Clotfelter (1983), the empirical analysis of taxpayer compliance has attracted increasing attention. However, despite these efforts, our understanding of compliance remains surprisingly limited. The fundamental difficulty in empirical work is the absence of detailed and reliable information on individual compliance choices. By its very nature, people have an incentive to hide information on their evasion behavior, and this concealment makes empirical work quite difficult. Most work in the United States has used information from the Taxpayer Compliance Measurement Program (TCMP) of the Internal Revenue Service (IRS).[1] However, TCMP data have some well-recognized deficiencies,[2] and until recently the IRS has only made these data available to researchers in aggregate form for the year 1969.[3] Partly in response, other work has attempted to use various indirect measures of evasion, such as the discrepancy between income reported on tax returns and income in the national income accounts.[4] While creatively derived and used, such surrogate measures are necessarily aggregate and approximate. Still other research has used data generated from surveys of taxpayers.[5] Unfortunately, these surveys are subject to a number of methodological problems that make their data highly suspect.[6] The difficulties with the various data sources have meant that no one single study has been able to construct and use measures of tax, penalty, and audit variables.[7] It seems unlikely that the quality of field data will soon—if ever—improve.

There is, however, another source of information on the compliance choices of individuals: data from laboratory experiments. There is a growing literature that uses experimental methods to examine the compliance behavior of individuals.[8] However, these data have not been fully utilized in the estimation of individual responses to government policies. In particular, no work estimates individual responses to a broad range of fiscal variables.

This paper provides such an analysis. Data from a series of laboratory experiments are collected, and these data are then used to estimate the individual responses to various policy changes. Unlike measures of evasion in previous empirical work, those that emerge from the laboratory are accurate and unambiguous measures of individual noncompliance; these measures are also derived in a setting that controls explicitly for extraneous influences on individual behavior. There are reasons for caution in using these data (see below). However, the laboratory offers perhaps the only opportunity to investigate in a controlled environment individual responses to tax, penalty, and audit rate changes, as well as to other policies for which field data do not readily exist, such as the role of fiscal exchange in tax compliance.

The estimation results confirm some

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(although not all) of the theoretical predictions about compliance behavior, and compare qualitatively with other empirical work based on TCMP and IRS data. Taxpayer reporting increases with an increase in the audit rate and a decrease in the tax rate. Compliance also depends upon the use of tax payments; that is, compliance is greater when individuals receive something for their taxes, despite the incentive to free ride on the contributions of others. The response to an increase in the penalty rate is positive but small and not highly significant. The estimation results therefore indicate that many, although not all, government policies can have a significant impact on compliance.

Theories of Tax Compliance

The standard theory of tax compliance is based on the work of Allingham and Sandmo (1972).[9] An individual is assumed to have a fixed endowment of income I, and must choose the amount to report to the tax authorities. Declared income D is taxed at the rate t. Unreported income is not taxed, but the individual may be audited with probability p, at which point a fine f is imposed on each dollar of unpaid taxes. The individual chooses D to maximize the expected utility of the evasion gamble.

This framework suggests that there is a demand for declared income that depends upon I, t, p, and f:

$$D = D(I,t,p,f). \quad (1)$$

It is straightforward to show that D increases with an increase in the probability of detection or the penalty rate, while the impacts of t and I depend upon the individual's attitude toward risk. Nearly all empirical work is based on this general specification.

There are, however, other factors that seem likely to affect the compliance decision. An obvious and largely neglected factor is government expenditures. As emphasized by Cowell and Gordon (1988, p. 305), "[t]his seems a curious oversight, since while the government taketh away, it also giveth back, and the latter activity surely exerts some influence on evasion."[10] The demand for declared income may therefore be modified to reflect the individual's receipt of government expenditures G:

$$D = D(I,t,p,f,G). \quad (2)$$

Equations (1) and (2) form the basis of the empirical specification.[11]

Experimental Design

The basic design of the experiments is straightforward.[12] Subjects receive income, they pay taxes on income voluntarily reported, they face a probability of audit, and they pay a penalty on taxes not paid if they are detected cheating; they may also receive a public good that depends upon the tax payments of all individuals. Into this setting various policy changes are introduced, and compliance is compared in the absence and the presence of the policy.

In each session, student subjects are organized into three groups of five, and each subject is assigned an identification number. The subjects are told that the session will last an unknown number of rounds; the actual number is predetermined at 25. Each subject is given an initial endowment of 10 tokens, and is told that accumulated tokens will be redeemed in cash at the end of the experiment. At the beginning of a round, each person is given income of either 2.00, 2.25, 2.50, 2.75, or 3.00 tokens. The amount is randomly chosen by the computer, and is known only to the subject. The subject must then decide how much of this income to report. Reported income is taxed at an announced rate. Unreported income is not taxed; however, the subject faces some probability of audit, at which point all unreported income for the current and the previous four rounds will be detected and a fine equal to an announced multiple of unpaid taxes will be imposed. The audit selection is determined by the use of a bingo cage that contains one numbered ball for each subject plus additional balls that determine the overall probability of audit. After all individuals have reported

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their income and paid their taxes in a round, up to one subject is chosen for audit by randomly drawing one ball with replacement from the bingo cage. Balls are placed in and drawn from the cage in full view of the subjects. At the end of each round, subjects are shown their new balances.

Subjects are volunteers from undergraduate classes, and have no prior experience with the experiments.[13] Each session is replicated with three different groups of five subjects; all run concurrently in the laboratory. All sessions begin with five practice rounds, and any questions are then answered. The full experiment then begins. Each session lasts approximately one hour, and each subject typically earns between $15 and $25, with the amount depending on the subject's performance; the average cost of each session is $300. The experiments are conducted in a computerized laboratory equipped with 15 subject terminals and a monitor terminal. All subject entries are made on the terminals, and all calculations are performed by the computer.

Several sessions are conducted, and their features are summarized in Table 1. In session 1, the tax rate on declared income is 0.30, the probability of audit is 0.04, and the fine rate on unpaid taxes is 2; these parameter values are chosen to approximate the tax, audit, and fine rates actually faced by individuals.[14] Other sessions introduce changes in the tax, audit, and penalty rates, and in the use of the tax revenues.

Session 2 introduces a public good. The tax payments of all five group members are paid into a group fund, this fund is increased by a multiple of 2 to reflect the consumers' surplus of the public good, and the resulting amount is distributed in equal shares to the group members. The subjects are informed of their share of this group fund at the end of each round. All other fiscal parameters are unchanged from session 1. Note that any amounts collected from the audits are not added to the group fund.

Sessions 3 through 8 make individual changes in the tax, fine, and audit rates. In session 3 the tax rate is reduced from 0.30 to 0.10, and in session 4 the tax rate is increased to 0.50. In sessions 5 and 6 the fine rate equals 1 and 3, respectively. The audit rate is reduced from 0.04 to 0.02 in session 7, and is increased to 0.06 in session 8. All other parameters equal their levels in session 1. There is no public good is these sessions.[15]

This experimental design suggests that the amount of income that the individual will declare in each round of the session takes the form:

$$\begin{aligned} \text{DECLARED} &= b_0 + b_1\,\text{INCOME} \\ &+ b_2\,\text{TAXRATE} + b_3\,\text{FINERATE} \\ &+ b_4\,\text{AUDITRATE} + b_5\,\text{PUBLICGOOD} \\ &+ b_6\,\text{PUBLICGOOD} \times \text{GROUPFUND}, \end{aligned}$$

where DECLARED is the amount of income reported by the individual, INCOME is the subject's income in the round, TAXRATE is the tax rate in the session, FINERATE is the fine rate, AUDITRATE is the probability of detection, PUBLICGOOD is a dummy variable that equals 1 if a public good is provided and 0 otherwise, and GROUPFUND is the amount of the group fund received by each subject in the previous round. Also estimated is an equation that omits the two public good variables.[16] Because the dependent variable DECLARED is censored at 0, the equations are estimated using the Tobit maximum likelihood method.[17]

Estimation Results

Average compliance rates in the various sessions are summarized in Table 1, and estimation results are reported in Table 2. Note that average compliance rates in Table 1 are approximately 1/3, or roughly comparable to average compliance rates estimated by the IRS (1988) for those income return items that are not subject to source withholding or that require interpretations of complex regulations (e.g., tips, proprietors' income, and Schedule E income). Note also that the estimation results in Table 2 are largely the same in both specifications.

Consider first the impact of income on

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TABLE 1
EXPERIMENTAL DESIGN

Session	Tax Rate	Fine Rate	Audit Rate	Public Good?	Average Compliance Rate[1]
1	0.30	2.0	0.04	No	0.332
2	0.30	2.0	0.04	Yes	0.374
3	0.10	2.0	0.04	No	0.376
4	0.50	2.0	0.04	No	0.200
5	0.30	1.0	0.04	No	0.317
6	0.30	3.0	0.04	No	0.376
7	0.30	2.0	0.02	No	0.321
8	0.30	2.0	0.06	No	0.365

[1] The average compliance rate is calculated as the ratio of declared income to actual income, averaged across all subjects, groups, and rounds in a session.

declared income. Higher income leads to higher compliance in both specifications of the compliance equation. This result suggests that declared income is a normal good, with an income elasticity (0.651 to 0.726) that is significantly less than one.[18] This result is also similar to other empirical work (Witte and Woodbury, 1985; Dubin, Graetz, and Wilde, 1990), in which the income elasticity of reported income is significant and positive.[19].

Higher tax rates lead to significantly lower compliance, which is consistent with the notion that the payoff to successful evasion is greater when the tax rate is larger. Note that most theoretical models of tax compliance conclude that compliance actually increases with greater tax rates, since a higher tax rate lowers income and, with decreasing absolute risk aversion, the lower income increases the individual's willingness to take risks.[20] However, the result here is similar to that of Clotfelter (1983), who finds that unreported income increases with higher marginal tax rates. The tax rate elasticities equal roughly −.5, a result similar to Clotfelter (1983).[21]

Compliance increases with an increase in the fine rate; however, the coefficient on FINERATE is so small that the fine rate elasticity is virtually zero, and the coefficient is also not highly significant. This result is not surprising. Since the probability of detection is small, large responses to changes in the fine rate would require extreme degrees of risk aversion. A policy implication is that increasing penalties may not have a noticeable effect on compliance, unless the probability of detection is increased significantly.

Compliance varies directly with changes in the audit rate, as suggested by the theoretical literature and the empirical work of Witte and Woodbury (1985), Dubin and Wilde (1988), and Dubin, Graetz, and Wilde (1990). However, like their results, the elasticity is not large, and equals approximately 0.17.[22] Given the difficulties in raising the audit rate significantly, the scope for increased compliance via increased audits seems limited.

The public good enters the estimation in two ways: as a dummy variable for the presence of a public good, and as an in-

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TABLE 2
ESTIMATION RESULTS[1]

	Model 1		Model 2	
Independent Variable	Coefficient Estimate	Elasticity	Coefficient Estimate	Elasticity
CONSTANT	-0.1458 (0.49)	--	-0.0559 (0.19)	--
INCOME	0.3943 (4.31)	0.726	0.3579 (4.16)	0.651
TAXRATE	-2.3630 (7.75)	-0.522	-2.3710 (7.73)	-0.517
FINERATE	0.0252 (1.43)	0.037	0.0251 (1.42)	0.037
AUDITRATE	5.7502 (1.89)	0.169	5.7691 (1.89)	0.168
PUBLICGOOD	--	--	-0.4068 (1.59)	--
PUBLICGOOD X GROUPFUND	--	--	0.9153 (2.06)	0.045
Log-likelihood	-3709.4		-4252.1	
n	2625		3000	

[1] The dependent variable is DECLARED. Asymptotic t-statistics are in parenthesis. Elasticities are computed at the mean values of the variables.

teraction variable that measures the individual payoff from the provision of the public good. The coefficient on the presence of the public good is negative and weakly significant, while the coefficient on the payoff to the public good is positive and highly significant. The former result indicates the presence of some free riding behavior, as individuals attempt to enjoy the public good without contributing. The latter result represents a Cournot response to the group fund (Cornes and Sandler, 1986); that is, individuals increase their compliance when they know that others are also contributing. Programs that make individuals aware of the benefits financed by their tax payments may be one tool for generating greater compliance.

Note finally that the data offer the opportunity to examine the potential bias—if any—that may be present in existing empirical studies by their inability to include all relevant fiscal variables in their specifications. The results here suggest that the (unavoidable) omission of fiscal variables in previous empirical studies may not seriously affect the results of those studies. For example, the tax rate elasticity equals approximately $-.5$ even when the audit rate or the fine rate is deliberately excluded from the specification.

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Similarly, the audit rate elasticity does not vary significantly when the tax rate or the fine rate is omitted. Consequently, even though measures of some variables may not always be available, estimation based on those variables that are available may well give an accurate measure of the compliance impact of these variables.

Conclusions

This paper uses data from laboratory experiments to estimate the effects on compliance of the major fiscal instruments. The empirical results indicate that tax compliance increases in income and audit rates and decreases in tax rates. Compliance is also greater when the individuals perceive some benefits from a public good funded by their tax payments, while changes in fine rates appear to have little effect on tax compliance behavior.

There are reasons for caution in the use of and generalization from these estimates. They are based upon somewhat artificial laboratory experiments, they are derived from student subjects, they are generated from small samples, and the effects on compliance of the various policy variables are often not large. Still, it seems likely that the results can contribute to our understanding of the compliance puzzle. Detailed and reliable field data on individual compliance choices do not exist and are unlikely to be soon available. Further, there is much evidence that there is little difference between student and nonstudent responses (Plott 1987). Most importantly, there is now a large literature that argues convincingly that experimental methods can contribute significantly to policy debates, as long as some conditions are met: the payoffs to subjects must be salient, better subject decisions yield higher subject payoffs, decision costs must be commensurate with the payoffs, and the experimental setting must capture the essential properties of the naturally occurring environment that is the subject of investigation (Smith 1982; Plott 1987). These conditions are met here.

It therefore appears that there are additional policy instruments beyond the standard prescription of greater enforcement actions that government can enact to achieve its desired degree of compliance with the tax laws. In fact, some of these standard instruments—greater penalties—may be largely ineffective in increasing tax compliance. In short, government should pursue a range of approaches in its efforts to promote compliance.

ENDNOTES

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[1]See, for example, Clotfelter (1983), Witte and Woodbury (1985), and Dubin and Wilde (1988). Only Clotfelter (1983) has had access to individual TCMP data, which he uses to estimate the impact of the marginal tax rate on evasion; however, he does not include any measure of the probability of audit. Witte and Woodbury (1985) and Dubin and Wilde (1988) include the audit rate in a compliance regression, but do not include the marginal tax rate; they also are forced to use TCMP data aggregated to the three-digit zip code level.

[2]TCMP data consist of detailed line-by-line audits of roughly 50,000 randomly-selected individual tax returns conducted on a three-year cycle; these audits yield an IRS estimate of the taxpayer's "true" income, so that a measure of individual tax evasion can be calculated. However, the IRS audits are likely to miss substantial amounts of unreported income, nonfilers are not captured at all, "honest" errors are not identified, and only audit recommendations, not final adjustments, are included.

[3]See, however, Tauchen, Witte, and Beron (1989) and Feinstein (1991) for empirical analysis of individual TCMP data.

[4]See, for example, Crane and Nourzad (1986) and Dubin, Graetz, and Wilde (1990).

[5]This literature is quite large. See, for example, Westat, Inc. (1980), Mason and Calvin (1984), and Yankelovich, Skelly, and White, Inc. (1984); see also the discussion in Roth, Scholz, and Witte (1989).

[6]For example, individuals may not remember their reporting decisions, they may not respond at all if they feel threatened by the sensitive nature of the questions, or they may not respond truthfully. See Klepper and Nagin (1989) for a more detailed discussion of survey methods in compliance research.

[7]There are also econometric problems with the existing empirical literature, especially in the treatment and specification of the audit variables. However, see Alm, Bahl, and Murray (1990) for empirical work on Jamaica that overcomes many of these data and econometric problems.

[8]See, for example, Friedland, Maital, and Rutenberg (1978), Spicer and Becker (1980), Becker, Buchner, and Sleeking (1987), Alm, Jackson, and McKee (forthcoming), and Alm, McClelland, and Schulze (forthcoming).

[9]See Cowell (1990) for an extensive discussion of the theoretical literature.

[10]In a related vein, there is an enormous literature

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on the voluntary provision of public goods, which suggests that individuals will often contribute to a public good even when there are no sanctions imposed on free-riders. See Dawes and Thaler (1988) for a brief discussion of this literature.

[11]There are, of course, other factors that affect the compliance decision: uncertainty about government policies, overweighting of low audit probabilities, the game-theoretic nature of the audit process, the role of tax practitioners, the influence of social norms, the process that determines how tax revenues are to be spent, and so on. See Roth, Scholz, and Witte (1989) for a discussion of some of these factors.

[12]The instructions are available upon request.

[13]The use of student subjects is nearly universal in experimental economics. Availability and cost play a role in this practice. More importantly, properly designed experiments evaluate individual decisionmaking, and there is no reason to doubt that students have access to the same cognitive processes as nonstudents. Studies that compare student and nonstudent responses have generally found no significant differences (Plott 1987).

[14]A penalty rate of 2 implies that subjects will pay back taxes plus a penalty equal to back taxes if caught cheating; the actual (maximum) penalty imposed on fraudulent reporting is back taxes plus 75 percent of back taxes. Similarly, the actual federal government audit rate has fallen in recent years to roughly 1 percent (Dubin, Graetz, and Wilde 1990). A tax rate of 30 percent in the experiments is comparable to that now faced by individuals in the top marginal tax bracket; it is also roughly in the middle of the tax rate schedules in place prior to the 1986 Tax Reform Act. For example, marginal tax rates ranged from 14 to 70 percent during most of the 1970s, and from 11 to 50 percent during much of the 1980s (Pechman 1987).

[15]Sessions 5, 6, 7, and 8 are discussed in more detail in Alm, Jackson, and McKee (forthcoming).

[16]Session 2 is omitted from this estimation.

[17]Numerous other specifications have also been estimated. For example, the dependent variable has been redefined as the amount of unreported income, the ratio of reported income to true income, and the ratio of unreported income to true income. The results are largely unaffected. It would be desirable to include sociodemographic variables (Beron, Tauchen, and Witte, 1989); however, the experimental protocol does not allow the collection of such information.

[18]The elasticities are computed at the mean values of the variables, following the procedure suggested by McDonald and Moffitt (1980). Note that the fraction of the mean total response due to the response above the limit value is .43 in both specifications.

[19]Witte and Woodbury (1985) report income elasticities that vary from 0.02 to 0.11, while Dubin, Graetz, and Wilde (1990) estimate income elasticities that equal unity.

[20]See, for example, Allingham and Sandmo (1972).

[21]Note that the tax rate elasticities here are for reported income; these estimates imply an unreported income-tax rate elasticity of roughly one-fourth. Clotfelter (1983) estimates the impact of marginal tax rates on unreported income, and finds elasticities that vary from .5 to 3.0.

[22]For example, the results in Dubin, Graetz, and Wilde (1990) imply a reported tax liability-audit rate elasticity of 0 08 to 0.18, and an assessed tax liability-audit rate elasticity of 0.09 to 0.19.

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